

Goal-Directed Navigation with Velocity Control

From Artificial Potential Fields (APF) to Control Barrier Functions (CBF)

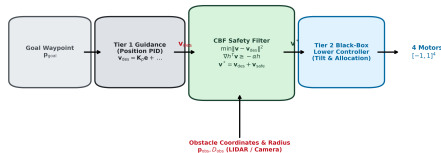
Autonomous Aerial Systems Course

Department of Robotics & Intelligent Systems

Introductory Lecture

Problem Formulation: High-Level Velocity as Control Input

Hierarchical Drone Control with Control Barrier Function Safety Shield



Tier 1: Kinematic Guidance ($\dot{\mathbf{p}} = \mathbf{v}_{cmd}$)

- Treats drone as a single integrator in $\mathbb{R}^3$.
- High-level controller outputs **velocity command** $\mathbf{v}_{cmd}$.
- Objective: Navigate from $\mathbf{p}_0$ to $\mathbf{p}_{goal}$ while avoiding obstacles at $\mathbf{p}_{obs}$.

Tier 2: 6-DOF Dynamic Tracking

- Fast inner attitude loop (PID / SE(3)).
- Tilts drone (pitch θ , roll ϕ) to track $\mathbf{v}_{cmd}$.
- Allocates motor thrusts $\Omega_{1,2,3,4} \in [-1, 1]^4$.

The Core Question: How do we compute $\mathbf{v}_{cmd}$? We explore **two major paradigms**: APF vs. CBF.

Paradigm 1: Artificial Potential Fields (APF)

Classic approach (Khatib 1986): Robot rolls downhill on an energy landscape $U(\mathbf{p}) = U_{\text{att}}(\mathbf{p}) + U_{\text{rep}}(\mathbf{p})$:

$$\mathbf{v}_{\text{cmd}}^{\text{APF}} = -\nabla U(\mathbf{p}) = \mathbf{v}_{\text{att}}(\mathbf{p}) + \mathbf{v}_{\text{rep}}(\mathbf{p})$$

Velocity Field Components

Goal Attraction (pulls toward target):

$$\mathbf{v}_{\text{att}}(\mathbf{p}) = -k_{\text{att}}(\mathbf{p} - \mathbf{p}_{\text{goal}})$$

Obstacle Repulsion (pushes outward within D_{obs}):

$$\mathbf{v}_{\text{rep}}(\mathbf{p}) = k_{\text{rep}} \left(\frac{1}{d} - \frac{1}{D_{\text{obs}}} \right) \frac{1}{d^2} \hat{\mathbf{n}}$$

where $d = \|\mathbf{p} - \mathbf{p}_{\text{obs}}\|$ and $\hat{\mathbf{n}} = \frac{\mathbf{p} - \mathbf{p}_{\text{obs}}}{d}$.

The Appeal of APF

- Very simple, closed-form vector sum.
- Intuitive physical analogy (charged particle attracted to goal and repelled by charges).
- Taught in robotics textbooks for 35+ years!

The Hidden Trap

APF **blends** goal tracking and safety into an unconstrained vector sum. This causes **four fatal failure modes** in practice!

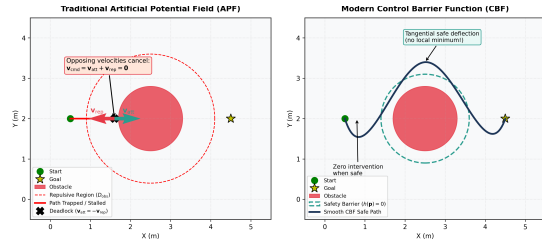
APF Failure Modes 1 & 2: Stalling and Goal Inaccessibility

Failure 1: Collinear Deadlock (Local Minimum)

- When start, obstacle, and goal align on a line:
- $\mathbf{v}_{att}$ pulls forward; $\mathbf{v}_{rep}$ pushes directly backward.
- At distance d^* , $\|\mathbf{v}_{att}\| = \|\mathbf{v}_{rep}\| \Rightarrow \mathbf{v}_{cmd} = \mathbf{0}$.
- The drone permanently freezes in mid-air!

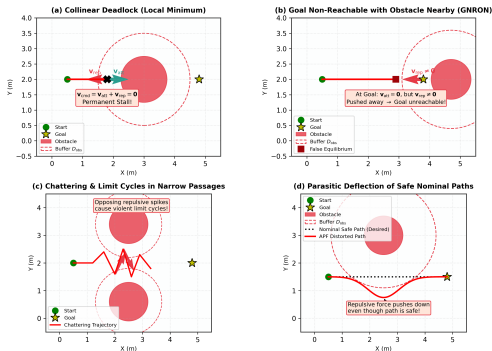
Failure 2: Goal Non-Reachable (GNRON)

- If an obstacle is close to the goal ($d(\mathbf{p}_{goal}) < D_{obs}$):
- At the goal: $\mathbf{v}_{att} = \mathbf{0}$, but $\mathbf{v}_{rep} \neq \mathbf{0}$!
- Robot is repelled away from its destination.



Vector Cancellation: At the local minimum, opposing velocities cancel out ($\mathbf{v}_{cmd} = \mathbf{0}$).

APF Failure Modes 3 & 4: Chattering and Parasitic Deflection



Failure 3: Chattering in Narrow Corridors

Repulsion scales as $1/d^3$. Near walls, explosive velocity spikes cause the inner attitude loop to oscillate violently and destabilize.

Failure 4: Parasitic Path Distortion

APF pushes radially even when the drone is flying **parallel to** or **away from** the obstacle, distorting collision-free paths.

The Paradigm Shift: From Vector Sum to Safety Filter

Why APF Breaks Down

APF attempts to solve two opposing goals simultaneously by **mixing them into one vector**:

$$\mathbf{v} = \mathbf{v}_{\text{att}} + \mathbf{v}_{\text{rep}}$$

This force fight corrupts nominal tracking and creates spurious zero-velocity equilibria.

The Co-Pilot Analogy

- The **student (PID)** steers toward the waypoint.
- The **instructor (CBF Filter)** sits quietly with hands off the controls.
- If and only if the student is about to clip an obstacle does the instructor nudge the stick just enough to remain safe!

The CBF Solution: Decoupling

- 1 **Goal Tracker**: Computes nominal velocity $\mathbf{v}_{\text{des}}$.
- 2 **Safety Filter**: Defines safe velocity set $\mathcal{K}(\mathbf{p})$.
- 3 **Minimal Intervention**: Projects $\mathbf{v}_{\text{des}} \rightarrow \mathbf{v}^*$ only when danger is imminent!



Step 1: Quantifying Safety with Barrier Function $h(\mathbf{p})$

For a spherical obstacle at position $\mathbf{p}_{\text{obs}}$ with safety buffer D_{obs} :

$$h(\mathbf{p}) = \|\mathbf{p} - \mathbf{p}_{\text{obs}}\| - D_{\text{obs}}$$

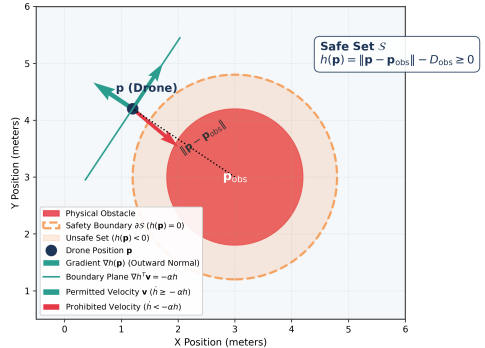
Three Fundamental Topological Sets

- **Safe Set ($\mathcal{S}$):** $h(\mathbf{p}) \geq 0$ (outside safety buffer)
- **Safety Boundary ($\partial\mathcal{S}$):** $h(\mathbf{p}) = 0$ (contact boundary)
- **Unsafe Set:** $h(\mathbf{p}) < 0$ (buffer penetrated!)

Outward Normal Vector (Gradient):

$$\nabla h(\mathbf{p}) = \frac{\mathbf{p} - \mathbf{p}_{\text{obs}}}{\|\mathbf{p} - \mathbf{p}_{\text{obs}}\|}, \quad \|\nabla h(\mathbf{p})\| = 1$$

Control Barrier Function (CBF) Geometry & Forward Invariance



Outward gradient $\nabla h(\mathbf{p})$ defines the barrier boundary and permitted velocity half-space.

Step 2: The Barrier Condition: $\dot{h} \geq -\alpha h(\mathbf{p})$

How does safety margin change as the drone moves with velocity $\mathbf{v} = \dot{\mathbf{p}}$?

$$\dot{h}(\mathbf{p}, \mathbf{v}) = \nabla h(\mathbf{p})^T \mathbf{v}$$

The Control Barrier Inequality (Ames et al.)

$$\dot{h}(\mathbf{p}, \mathbf{v}) \geq -\alpha h(\mathbf{p}) \iff \nabla h(\mathbf{p})^T \mathbf{v} \geq -\alpha h(\mathbf{p})$$

Physical Meaning (Dynamic Speed Limit):

- **Far away** ($h \gg 0$): Allowed to fly toward obstacle fast.
- **Approaching** ($h \rightarrow 0$): Inward speed smoothly throttled.
- **At boundary** ($h = 0$): Inward motion strictly forbidden ($\nabla h^T \mathbf{v} \geq 0$).

Everyday Analogy: Smooth Braking

- A bad driver slams brakes at the last second.
- A smooth driver brakes gradually: speed limit is proportional to distance remaining!
- That is exactly what $\dot{h} \geq -\alpha h$ enforces.

Tuning α (Class- $\mathcal{K}$ Gain)

- **Small α (0.5)**: Early, gentle, conservative.
- **Large α (2.5)**: Tighter, agile, aggressive turn.

The CBF-QP Safety Filter: Exact Closed-Form Solution

We formulate safety filtering as a Quadratic Program:

$$\mathbf{v}^* = \arg \min_{\mathbf{v} \in \mathbb{R}^3} \frac{1}{2} \|\mathbf{v} - \mathbf{v}_{\text{des}}\|^2 \quad \text{s.t.} \quad \nabla h^T \mathbf{v} \geq -\alpha h$$

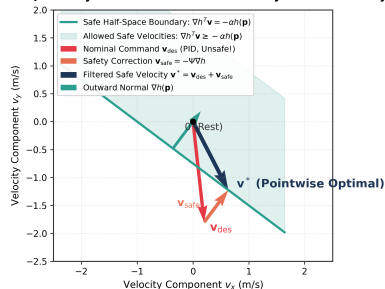
Analytical KKT Solution

Define the safety violation scalar Ψ :

$$\Psi = \nabla h(\mathbf{p})^T \mathbf{v}_{\text{des}} + \alpha h(\mathbf{p})$$

- If $\Psi \geq 0$: $\mathbf{v}^* = \mathbf{v}_{\text{des}}$ (**Zero intervention!**)
 - If $\Psi < 0$: $\mathbf{v}^* = \mathbf{v}_{\text{des}} - \Psi \nabla h(\mathbf{p})$
-
- One dot product + vector subtraction ($< 1 \mu\text{s}$)!
 - No numerical QP solver needed on flight MCU.
 - Guaranteed minimal distortion $\|\mathbf{v}^* - \mathbf{v}_{\text{des}}\|$.

CBF-QP Safety Filter: Minimal-Invasive Projection Geometry



Geometric Projection: When $\Psi < 0$, $\mathbf{v}_{\text{des}}$ is projected orthogonally along ∇h onto the safe boundary.

Why CBFs Overcome All Four APF Failure Modes

1. No Collinear Deadlocks (Traps)

- APF: Opposing velocities cancel out ($\mathbf{v} = \mathbf{0}$).
- **CBF**: Only normal inward component restricted. Tangential plane is **unconstrained** $\implies$ smooth slide!

2. No GNRON (Goal Near Obstacle)

- APF: $\mathbf{v}_{\text{rep}} \neq \mathbf{0}$ at goal pushes drone away.
- **CBF**: As $\mathbf{p} \rightarrow \mathbf{p}_{\text{goal}}$, $\mathbf{v}_{\text{des}} \rightarrow \mathbf{0} \implies \nabla h^T \mathbf{0} \geq -\alpha h$. Drone stops exactly at the goal!

3. No Chattering / Limit Cycles

- APF: $1/d^3$ gradient explosion triggers attitude oscillations.
- **CBF**: Linear velocity constraint produces smooth, bounded deflections.

4. Zero Parasitic Deflection

- APF: Repels even when flying parallel/away.
- **CBF**: Flying parallel or away satisfies $\Psi \geq 0 \implies \mathbf{v}^* = \mathbf{v}_{\text{des}}$ (100% unhampered tracking!).

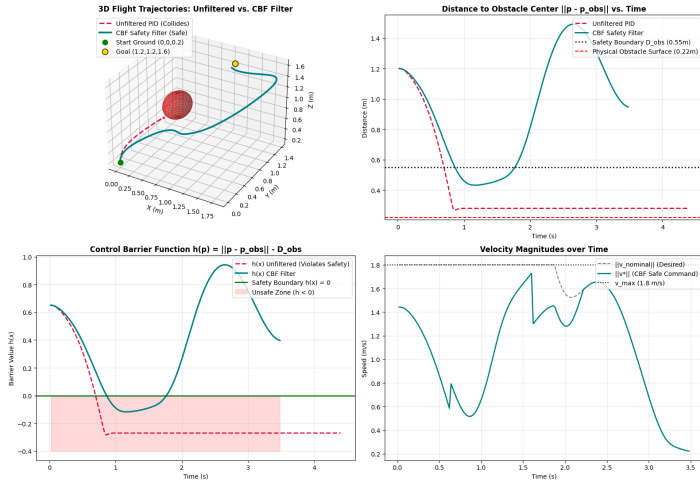
Head-to-Head Analytical Comparison: APF vs. CBF

Feature	Artificial Potential Fields (APF)	Control Barrier Functions (CBF)
Core Paradigm	Vector Sum: $\mathbf{v}_{att} + \mathbf{v}_{rep}$	Constrained Projection: $\arg \min \ \mathbf{v} - \mathbf{v}_{des}\ ^2$
Safety Formulation	Soft energy penalty U_{rep}	Hard forward-invariance set constraint
Nominal Preservation	Corrupted inside D_{obs}	Preserved exactly whenever safe ($\mathbf{v}^* = \mathbf{v}_{des}$)
Collinear Deadlock	Severe (permanent stall)	Eliminated (tangential slide)
Goal Near Obstacle	Fails (equilibrium shifted)	Guaranteed exact arrival
Chattering Oscillations	High ($1/d^3$ explosive gradient)	None (smooth linear boundary)
Computational Cost	$\mathcal{O}(1)$ vector sum	$\mathcal{O}(1)$ closed-form KKT formula
Formal Guarantees	None	Provable forward invariance $\mathbf{p}(t) \in \mathcal{S}$

Summary Takeaway

CBF retains the analytical speed of APF ($\mathcal{O}(1)$ closed-form) while eliminating all topological traps and providing formal set-invariance safety guarantees!

Flight Verification in 6-DOF MuJoCo Simulation



Top: 3D flight paths (obstacle collision vs. smooth circumnavigation) & obstacle clearance. **Bottom:** Barrier $h(t) \geq 0$ invariant; smooth filtered velocity command $\mathbf{v}^*$.

Telemetry Verification: The Numbers Don't Lie

Metric	Unfiltered PID (Exp A)	CBF Safety Filter (Exp B)
Obstacle Center Position	[0.6, 0.6, 1.05] m	[0.6, 0.6, 1.05] m
Obstacle Radius (R_{obs})	0.22 m	0.22 m
Configured Buffer (D_{obs})	0.55 m	0.55 m
Minimum Center Distance	0.266 m	0.432 m
Minimum Surface Clearance	0.046 m	0.212 m
Minimum Barrier Value $h(t)$	-0.284	≥ 0
Propeller Blade Collision?	YES (CRASH!)	NO (Zero touch!)
Goal Waypoint Reached?	Failed	Success (Step 173)

Key Takeaway

The unfiltered PID breached the blade radius ($0.266 \text{ m} < 0.39 \text{ m}$), shearing propellers off the obstacle. The CBF-filtered drone maintained $> 0.21 \text{ m}$ blade clearance and completed the mission.

Engineering Lessons for Autonomous Drone Roboticians

1. Pad Buffer for Tilt Inertia

Physical drones cannot change velocity instantaneously—they must tilt ($\Delta t_{\text{tilt}} \approx 0.15 \text{ s}$). Always set:

$$D_{\text{obs}} \geq R_{\text{obs}} + R_{\text{drone}} + v_{\text{max}} \Delta t_{\text{tilt}}$$

2. Break Symmetry on Head-On Encounters

If flying directly head-on ($-\nabla h$), inject an infinitesimal perturbation $\epsilon = \epsilon(\nabla h \times \hat{z})$ to guide the drone around the obstacle effortlessly!

Summary of What We Learned

- 1 **Kinematic Velocity Control:** Decouples high-level guidance from complex 6-DOF drone dynamics.
- 2 **APF Limits:** Adding velocities causes local minima stalls, GNRON, and violent chattering.
- 3 **CBF Breakthrough:** Safety filtering via set invariance ($\dot{h} \geq -\alpha h$) preserves nominal tracking and guarantees safety.
- 4 **Closed-Form QP:** Solves in $< 1 \mu\text{s}$ onboard without numerical solvers!

Thank you! Questions?